K L Deemed to be University	Department of computer science and Engineering -1 -- KLBCH Course Handout 2026-2027, Even Sem

Course Title	**: GENERATIVE AI AND AI PLATFORM ENGINEERING**
Course Code	**: 25CS2207E**
L-T-P-S Structure	**: 2-0-0-8**
Pre-requisite	: 25SC2107E-MACHINE LEARNING
Credits	**: 4**
Course Coordinator	**: DHARAMSOTU BHEEKYA**
Team of Instructors	: Dyuti Banerjee , Dyuti Banerjee , Gunturi Raghavendra, Gunturi Raghavendra, ASIF SHAIK, ASIF SHAIK, khaja sk, khaja sk, Anitha Chekuri, Nirmalajyothi Narisetty, Nirmalajyothi Narisetty, LakshmiPrasanna Yeluri , LakshmiPrasanna Yeluri , Swathi D, KIRAN KUMAR ANUMANDLA, KIRAN KUMAR ANUMANDLA, Chandusha kanda, Purushotham Muniganti, Purushotham Muniganti, Jyothsna Datti, Jyothsna Datti, Rahul LR, Rahul LR, I V SAI HARITHA, I V SAI HARITHA, Kavitha Matta, Kavitha Matta, Gayatri Mantri, Kotte Anusha, Kotte Anusha, DHARAMSOTU BHEEKYA, DHARAMSOTU BHEEKYA, pavani Chilumula, Pratima Depa, Divya Priya Degala, MEKALA SANJEEV KUMAR, Anu Ragavi V, Anu Ragavi V
Teaching Associates	:
Syllabus :	Module I – Generative AI and Tokenization covers the Generative AI landscape, AI, ML, DL, NLP and GenAI, generative versus discriminative tasks, modalities, LLMs, foundation models, next-token prediction, open-weight and closed models, model capabilities, hallucinations, knowledge cutoffs, pricing, rate limits, failure rates, tokenization, tokens, IDs, vocabulary, BPE, WordPiece, SentencePiece, English, Telugu and Hindi tokenization, and token economics, including input/output token counts, context limits and cost estimation. Module II – Foundations of Generative Models focuses on autoregressive generation, next-token probability distributions, token embeddings, positional information, attention mechanisms involving query, key and value, dot-product scores, softmax, self-attention, multi-head attention, Transformer architecture, feed-forward networks, residual connections, layer normalisation, encoder-only (BERT), decoder-only (GPT) and encoder–decoder (T5) models, decoding strategies such as temperature, top-k and top-p, scaling, pre-training, fine-tuning, alignment concepts and model limitations. Module III – Embeddings and Vector Stores: The RAG Pattern deals with dense semantic embeddings, sparse representations such as TF-IDF and BM25, sentence-transformers, document ingestion, cleaning, metadata management, chunking with overlap, chunk-size selection, MongoDB as the system of record, Chroma as the vector store, HNSW and IVF indexing, recall–latency trade-offs, Atlas Vector Search, pgvector, the end-to-end Retrieval-Augmented Generation (RAG) pipeline, retrieval, context construction, grounded generation, citations, failure cases and retrieval evaluation using Hit Rate@k, MRR and NDCG. Module IV – Prompt Engineering and In-Context Learning covers system and user prompts, instructions, context, output specifications, prompt versioning, zero-shot and few-shot learning, example selection, role prompting, chain-of-thought awareness, responsible usage, structured outputs using Pydantic schemas, validation, retries, safe fallbacks, controlled tool calling, validated arguments, allow-lists, evaluation rubrics, regression testing, LLM-as-a-judge, human agreement, prompt injection, information leakage and quality regression. Module V – Fine-Tuning and Adaptation explores fine-tuning versus prompting versus RAG, domain-specific, style-based and narrow-task adaptation, build-versus-buy decisions, full fine-tuning, Parameter-Efficient Fine-Tuning (PEFT), LoRA, QLoRA, adapters, quantisation, instruction dataset design, data quality, documentation, supervised fine-tuning, alignment methods including RLHF/PPO, DPO and RLAIF, and the trade-offs

	involving quality, effort, memory, latency and cost. Module VI – AI Platform Engineering focuses on AI application architecture, including UI, API, orchestration, retrieval, model, post-processing and safety layers, clean architecture, 12-factor configuration, model selection based on task, language, cost, latency and privacy, FastAPI microservices with Pydantic contracts, React + Vite using TypeScript and JavaScript, Tailwind CSS with MUI and Bootstrap, operational tracing using Langfuse or LangSmith, cost control, graceful degradation, guardrails, human-in-the-loop mechanisms, kill switches, Docker and docker-compose, hardened images, GitHub Actions with Trivy and tfsec/checkov, Terraform-based AWS infrastructure using VPC, IAM, ECR and EC2, and MCP as an enrichment mechanism.
Text Books :	[1] C. Huyen, AI Engineering: Building Applications with Foundation Models. Sebastopol, CA, USA: O'Reilly Media, 2025. [2] J. Alammar and M. Grootendorst, Hands-On Large Language Models: Language Understanding and Generation. Sebastopol, CA, USA: O'Reilly Media, 2024. [3] S. Raschka, Build a Large Language Model (From Scratch). Shelter Island, NY, USA: Manning Publications, 2024. [4] J. Berryman and A. Ziegler, Prompt Engineering for LLMs. Sebastopol, CA, USA: O'Reilly Media, 2024. [5] P. Iusztin and M. Labonne, LLM Engineer's Handbook. Birmingham, U.K.: Packt Publishing, 2024.
Reference Books :	[1] M. Richards and N. Ford, Fundamentals of Software Architecture. Sebastopol, CA, USA: O'Reilly Media, 2020. [2] Y. Brikman, Terraform: Up and Running, 3rd ed. Sebastopol, CA, USA: O'Reilly Media, 2022. [3] S. Bradshaw, E. Brazil, and K. Chodorow, MongoDB: The Definitive Guide, 3rd ed. Sebastopol, CA, USA: O'Reilly Media, 2019.
Web Links :	1. https://huggingface.co/learn/llm-course/chapter1/1 2. https://developers.openai.com/api/docs 3. https://docs.langchain.com/oss/python/langchain/overview?utm_source=chatgpt.com 4. https://fastapi.tiangolo.com/?utm_source=chatgpt.com
MOOCS :	1. Generative AI with Large Language Models, https://www.coursera.org/learn/generative-ai-with-llms 2. Generative AI for Every one, https://www.coursera.org/organizations/kluniversity/learn/generative-ai-for-everyone 3. Building Generative AI-Powered Applications with Python - https://www.coursera.org/organizations/kluniversity/learn/building-gen-ai-powered-applications
Course Rationale :	Generative AI and AI Platform Engineering (GALE) places students inside the way production AI applications are now built: a foundation model treated as a dependency, wrapped in retrieval, prompts, evaluation, security and cloud delivery. The PBL track is OUTSIDE-IN. On day one students see a working, cited course assistant — React + Vite front end, FastAPI microservices, MongoDB and Chroma, deployed on AWS — and over the trimester they rebuild every layer that powers it. Every module ends in a measured release. A golden question set, written in Module I, is re-run after every module, so each design choice (generation defaults, chunk size, prompt version, fine-tune or not) is backed by numbers and recorded as an architecture decision. The project moves from v0.1 (a logged LLM client) to v1.0 (a secure, governed, personalised application on AWS), and the v0.1 → v1.0 record is the final evidence. Pre-requisite: MLNG. The course builds directly on machine-learning fundamentals and prepares students for AI engineer, LLM application developer, MLOps/LLMOps and AI platform roles.
Course Objectives :	The course enables students to design, build, evaluate and operate Generative AI applications to industry standard. Through an outside-in, project-based approach, students start from a working AI application and progressively reconstruct it: tokenization and cost, Transformer generation, retrieval-augmented generation, prompt engineering with structured outputs and tool calling, fine-tuning and adaptation, and platform engineering with tracing,

	guardrails, CI and Infrastructure as Code. By the end of the course, students will be able to ship a measured, secure and governed AI application and defend each design decision with evidence.
Industry Specific Tools :	Hugging Face, OpenAI API, Ollama, Google Colab, LangChain, Chroma, MongoDB Atlas, FastAPI, React, Docker, GitHub Actions, Terraform, AWS.
Sector Skill Council :	IT-ITeS Sector Skill Council – NASSCOM (SSC NASSCOM). https://nqr.gov.in/qualifications/14770?utm_source=chatgpt.com

COURSE OUTCOMES (COs):

CO NO	Course Outcome (CO)	PO/PSO	Blooms Taxonomy Level (BTL)
CO1	Apply Generative AI and tokenization fundamentals to build a provider-agnostic LLM client that logs tokens, latency and cost, quantify tokenization cost across English and Indian languages, and establish the golden question set every later module re-runs	PO2,PO3,PO5,PSO1,PO1	3
CO2	Analyse how Transformer models generate text by comparing encoder, decoder and encoder–decoder families, inspecting attention and sampling behaviour and classifying failures — then select generation defaults from measured evidence recorded as an architecture decision.	PSO1,PO10,PO5,PO4,PO2,PO1	4
CO3	Create a cited RAG service over course material — MongoDB as the system of record, Chroma as the vector store — that measures retrieval with Hit Rate@k and MRR and declines questions its sources do not support.	PO1,PO2,PO3,PO5,PSO1,PSO2	6
CO4	Create versioned, schema-validated prompts with controlled tool calling, compare zero- and few-shot designs with numbers, and defend them with an injection-and-leakage suite and judge-calibrated regression tests in CI.	PO3,PSO1,PO9,PO5,PO2,PO8	6
CO5	Evaluate fine-tuning against prompting and retrieval baselines on one test set — accuracy, latency and cost — and ship the evidence-backed choice as a learned router with a documented dataset and decision record.	PO2,PO4,PO5,PO11,PSO1	5
CO6	Create and operate a secure, governed, hyper-personalised AI application — React + Vite, FastAPI microservices, MongoDB and Chroma — with tracing, CI and an ethics layer, deployed on AWS through Docker and Terraform with a measured v0.1 → v1.0 record.	PSO2,PSO1,PO11,PO10,PO9,PO8,PO6,PO5,PO3	6

COURSE OUTCOME INDICATORS (COIs)::

Outcome No.	Highest BTL	COI-1	COI-2	COI-3	COI-4	COI-5	COI-6
CO1	3	**Btl-1** Recall the AI → ML → DL → NLP → GenAI hierarchy, model types (open-weight, closed, foundation) and the vocabulary of tokens, IDs and context limits.	**Btl-2** Explain BPE, WordPiece and SentencePiece tokenization, why Telugu and Hindi cost more tokens than English, and how hallucination, knowledge cutoffs, rate limits and failure rates make a model a dependency.	**Btl-3** Apply token-cost arithmetic to budget a feature, and build a provider-agnostic LLM client that logs tokens, latency and cost against the golden question set.			
CO2	4	**Btl-1** Recall the parts of a Transformer block and the three model families (BERT, GPT, T5).	**Btl-2** Explain autoregressive generation, embeddings, positional information and scaled dot-product, self- and multi-head attention.	**Btl-3** Apply temperature, top-k and top-p decoding and inspect real attention weights across encoder, decoder and encoder–decoder models.	**Btl-4** Analyse sampling behaviour, classify generation failures and select generation defaults from measured evidence, recorded as an architecture decision.		
CO3	6		**Btl-2** Explain dense versus sparse (TF-IDF, BM25) representations and the recall–latency trade-off of HNSW and IVF indexes.	**Btl-3** Apply an ingestion pipeline — cleaning, metadata, chunking with overlap — storing documents in MongoDB and embeddings in Chroma.		**Btl-5** Evaluate retrieval on its own with Hit Rate@k and MRR (NDCG awareness) and choose chunk size and k by measurement.	**Btl-6** Create a cited RAG service that grounds answers in retrieved sources and declines questions the sources do not support.
CO4	6		**Btl-2** Explain the prompt as a versioned component: system and user prompts, instructions, context and	**Btl-3** Apply zero-shot, few-shot and role prompting and Pydantic-validated structured outputs with		**Btl-5** Evaluate retrieval on its own with Hit Rate@k and MRR (NDCG awareness) and choose chunk size	**Btl-6** Create a cited RAG service that grounds answers in retrieved sources and declines questions the

Outcome No.	Highest BTL	COI-1	COI-2	COI-3	COI-4	COI-5	COI-6
			output specification.	retry and safe fallback.		and k by measurement.	sources do not support.
CO5	5		**Btl-2** Explain the prompt as a versioned component: system and user prompts, instructions, context and output specification.	**Btl-3** Apply zero-shot, few-shot and role prompting and Pydantic-validated structured outputs with retry and safe fallback.	**Btl-4** Analyse alignment methods (RLHF/PPO, DPO, RLAIF) and the quality, effort, memory, latency and cost trade-offs of adaptation.	**Btl-5** Evaluate fine-tuning against prompting and retrieval baselines on one test set and ship the choice as a learned router with a decision record.	
CO6	6		**Btl-2** Explain the layers of an AI application, clean architecture, 12-factor configuration and model selection by task, language, cost, latency and privacy.	**Btl-3** Apply FastAPI microservices with Pydantic contracts and a React + Vite front end over MongoDB and Chroma.		**Btl-5** Evaluate the running system through per-request traces, cost control, graceful degradation, guardrails and human-in-the-loop review.	**Btl-6** Create and deploy a secure, governed application on AWS with Docker, GitHub Actions security scans and Terraform, documented as a v0.1 → v1.0 record.

PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES (POs/PSOs)

Po No.	Program Outcome

Lecture Course DELIVERY Plan:

Sess.No.	CO	COI	Topic	Book No[CH No] [Page No]	Teaching-Learning Methods	EvaluationComponents
1	CO1	COI-2	Course handout; outside-in project and golden question set	nan	LTC,PPT,Talk	End Semester Exam,Mid-Term Examination (Descriptive),MOOCs Completion & Certification
2	CO1	COI-2	AI → ML → DL → NLP → GenAI; generative versus discriminative tasks	T1 [CH1], T2 [CH1]	LTC,PPT,Talk	End Semester Exam,Mid-Term Examination (Descriptive),MOOCs Completion & Certification

Sess.No.	CO	COI	Topic	Book No[CH No] [Page No]	Teaching-Learning Methods	EvaluationComponents
3	CO1	COI-2	Modalities, LLMs, foundation models, next-token prediction; models as dependencies	T1 [CH1, CH2]	LTC,PPT,Talk	End Semester Exam,Mid-Term Examination (Descriptive),MOOCs Completion & Certification
4	CO1	COI-2	Tokens, IDs, vocabulary; BPE, WordPiece and SentencePiece	T2 [CH2], T3 [CH2]	LTC,PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Mid-Term Examination (Descriptive),MOOCs Completion & Certification
5	CO1	COI-3	English versus Telugu and Hindi tokenization; token economics and cost budgeting	T1 [CH2], T2 [CH2]	LTC,PPT,Talk	End Semester Exam,Mid-Term Examination (Descriptive),MOOCs Completion & Certification
6	CO2	COI-2	Autoregressive generation; next-token probabilities, embeddings and positional information	T3 [CH2], T2 [CH3]	LTC,PPT,Talk	End Semester Exam,Mid-Term Examination (Descriptive),MOOCs Completion & Certification
7	CO2	COI-2	Query, key and value; dot-product scores, softmax and self-/multi-head attention	T3 [CH3]	LTC,PPT,Talk	End Semester Exam,Mid-Term Examination (Descriptive),MOOCs Completion & Certification
8	CO2	COI-3	Transformer block; attention weights, feed-forward, residuals and layer normalisation	T3 [CH4], T2 [CH3]	LTC,PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Mid-Term Examination (Descriptive),MOOCs Completion & Certification
9	CO2	COI-3	Encoder-only BERT, decoder-only GPT and encoder–decoder T5	T2 [CH1, CH3]	LTC,PPT,Talk	End Semester Exam,Mid-Term Examination (Descriptive),MOOCs Completion & Certification
10	CO2	COI-4	Temperature, top-k, top-p; scaling, training, alignment and failure classification	T3 [CH4, CH5], T1 [CH2]	LTC,PPT,Talk	End Semester Exam,Mid-Term Examination (Descriptive),MOOCs Completion & Certification

Sess.No.	CO	COI	Topic	Book No[CH No] [Page No]	Teaching-Learning Methods	EvaluationComponents
11	CO3	COI-2	Dense embeddings versus sparse TF-IDF/BM25; sentence-transformers	T2 [CH2, CH8]	LTC,PPT,Talk	End Semester Exam,Mid-Term Examination (Descriptive),MOOCs Completion & Certification
12	CO3	COI-3	Document loading, cleaning, metadata and chunking with overlap	T1 [CH6], T5 [CH4]	LTC,PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,Mid-Term Examination (Descriptive),MOOCs Completion & Certification
13	CO3	COI-3	MongoDB system of record, Chroma vector store, HNSW/IVF and vector search	T2 [CH8], T5 [CH9]	LTC,PPT,Talk	End Semester Exam,Mid-Term Examination (Descriptive),MOOCs Completion & Certification
14	CO3	COI-5	Retrieval, context construction, grounded generation, citations and failure cases	T1 [CH3, CH6]	LTC,PPT,Talk	End Semester Exam,Mid-Term Examination (Descriptive),MOOCs Completion & Certification
15	CO3	COI-6	Hit Rate@k, MRR, NDCG awareness and unsupported-question refusal	T4 [CH1, CH5, CH6]	LTC,PPT,Talk	End Semester Exam,Mid-Term Examination (Descriptive),MOOCs Completion & Certification
16	CO4	COI-2	System/user prompts, instructions, context and versioned output specifications	T2 [CH6], T1 [CH5]	LTC,PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,MOOCs Completion & Certification
17	CO4	COI-3	Zero-shot, few-shot, example selection, role prompting and responsible use	T4 [CH8], T1 [CH6]	LTC,PPT,Talk	End Semester Exam,MOOCs Completion & Certification
18	CO4	COI-3	Pydantic structured outputs, validation, retry and safe fallback	T1 [CH3, CH4], T4 [CH10]	LTC,PPT,Talk	End Semester Exam,MOOCs Completion & Certification
19	CO4	COI-5	Controlled tool calling, validated arguments and allow-lists	T1 [CH7]	LTC,PPT,Talk	End Semester Exam,MOOCs

Sess.No.	CO	COI	Topic	Book No[CH No] [Page No]	Teaching-Learning Methods	EvaluationComponents
						Completion & Certification
20	CO4	COI-6	Rubrics, regression tests, LLM-as-a-judge, injection, leakage and quality regression	T1 [CH7], T3 [Appendix E], T2 [CH12]	LTC,PPT,Talk	End Semester Exam,MOOCs Completion & Certification
21	CO5	COI-2	Fine-tuning versus prompting versus RAG; adaptation and build-versus-buy	T1 [CH8], T5 [CH5], T3 [CH7]	LTC,PPT,Talk	End Semester Exam,MOOCs Completion & Certification
22	CO5	COI-3	Full fine-tuning versus PEFT; LoRA, QLoRA, adapters and quantisation	T5 [CH6], T2 [CH12]	LTC,PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,MOOCs Completion & Certification
23	CO5	COI-3	Instruction dataset design, quality, documentation and supervised fine-tuning	T1 [CH10], R1 [CH17, CH19]	LTC,PPT,Talk	End Semester Exam,MOOCs Completion & Certification
24	CO5	COI-4	RLHF/PPO, DPO and RLAIF awareness	R1 [CH17], T5 [CH10]	LTC,PPT,Talk	End Semester Exam,MOOCs Completion & Certification
25	CO5	COI-5	Quality, effort, memory, latency and cost; learned router and decision record	T1 [CH10], T5 [CH11]	LTC,PPT,Talk	End Semester Exam,MOOCs Completion & Certification
26	CO6	COI-2	AI application layers, clean architecture, 12-factor configuration and model selection	R2 [CH2, CH3, CH8], T5 [CH10, CH11]	LTC,PPT,Talk	End Semester Exam,MOOCs Completion & Certification
27	CO6	COI-3	FastAPI microservices with Pydantic contracts	R1 [CH17], T5 [CH10]	LTC,PPT,Talk	End Semester Exam,MOOCs Completion & Certification
28	CO6	COI-5	React + Vite, TypeScript/JavaScript, Tailwind CSS, MUI and Bootstrap	T1 [CH10], T5 [CH11]	LTC,PPT,Talk	AI-Enabled Adaptive Learning Engagement,End Semester Exam,MOOCs Completion & Certification

Sess.No.	CO	COI	Topic	Book No[CH No] [Page No]	Teaching-Learning Methods	EvaluationComponents
29	CO6	COI-5	Langfuse/LangSmith tracing, cost control, graceful degradation, guardrails and human review	T1 [CH10], T5 [CH11]	LTC,PPT,Talk	End Semester Exam,MOOCs Completion & Certification
30	CO6	COI-6	Docker, GitHub Actions security scans, Terraform on AWS and MCP enrichment	R2 [CH2, CH3, CH8], T5 [CH10, CH11]	LTC,PPT	End Semester Exam,MOOCs Completion & Certification

Lecture Session wise Teaching – Learning Plan

SESSION NUMBER : 1

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction to the subject	1	Talk	--- NOT APPLICABLE ---
20	Course handout: COs, evaluation, policies	2	Talk	--- NOT APPLICABLE ---
20	Live Talk of the finished course assistant; the golden question set	1	Talk	--- NOT APPLICABLE ---
5	Clarifications and course expectations	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 2

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	AI → ML → DL → NLP → GenAI	2	Talk	--- NOT APPLICABLE ---
20	Generative vs discriminative tasks with examples	2	PPT	--- NOT APPLICABLE ---
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 3

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	Modalities, LLMs, foundation models, next-token prediction	2	Talk	--- NOT APPLICABLE ---
20	Models as dependencies: open vs closed, hallucinations, cutoffs, price, rate limits, failure rates	2	PPT	Quiz/Test Questions
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 4

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	Tokens, IDs, vocabulary; BPE	2	Talk	One minute paper
20	WordPiece and SentencePiece through tokenizer libraries	2	Talk	--- NOT APPLICABLE ---
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 5

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	English vs Telugu and Hindi token counts	2	Talk	--- NOT APPLICABLE ---
20	Cost arithmetic: input/output, context and rate limits; budgeting a feature	3	Talk	Problem-Based Learning
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 6

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	Next-token probability distributions	2	Talk	--- NOT APPLICABLE ---
20	Token embeddings and positional information	2	PPT	--- NOT APPLICABLE ---
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 7

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	Query, key, value; dot-product scores and softmax	2	Talk	--- NOT APPLICABLE ---
20	Self- and multi-head attention	2	PPT	Quiz/Test Questions
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 8

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	The Transformer block: attention, feed-forward, residuals, layer norm	2	Talk	--- NOT APPLICABLE ---
20	Inspecting real attention weights	3	LTC	In-class Demonstrations
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 9

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	Encoder-only (BERT) and decoder-only (GPT)	2	Talk	--- NOT APPLICABLE ---
20	Encoder–decoder (T5); choosing a family for a task	3	PPT	--- NOT APPLICABLE ---
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 10

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	Temperature, top-k, top-p	3	Talk	--- NOT APPLICABLE ---
20	Scaling, pre-training, fine-tuning, alignment; classifying failures; writing an ADR	4	Talk	Inquiry Based Learning
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 11

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	Dense semantic embeddings	2	Talk	Quiz/Test Questions
20	Sparse TF-IDF and BM25; sentence-transformers	2	PPT	--- NOT APPLICABLE ---
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 12

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	Loading, cleaning and metadata	2	Talk	--- NOT APPLICABLE ---
20	Chunking with overlap; measuring chunk size	3	Talk	Case Study Based Learning
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 13

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	MongoDB as system of record; Chroma as vector store	2	Talk	--- NOT APPLICABLE ---
20	HNSW/IVF, recall–latency; Atlas Vector Search and pgvector	4	PPT	--- NOT APPLICABLE ---
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 14

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	Retrieval and context construction	4	Talk	--- NOT APPLICABLE ---
20	Grounded generation, citations, failure cases	5	Talk	Case Study Based Learning
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 15

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	Hit Rate@k and MRR	5	Talk	--- NOT APPLICABLE ---
20	NDCG awareness; declining unsupported questions	6	PPT	--- NOT APPLICABLE ---
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 16

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	System and user prompts, instructions, context	2	Talk	--- NOT APPLICABLE ---
20	Output specification; prompts versioned as files	3	PPT	--- NOT APPLICABLE ---
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 17

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	Zero-shot and few-shot; example selection	3	Talk	--- NOT APPLICABLE ---
20	Role prompting; chain-of-thought awareness and responsible use	3	Talk	--- NOT APPLICABLE ---
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 18

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	Pydantic schemas for structured output	3	Talk	Group Discussion
20	Validation, retry and safe fallback	3	Talk	--- NOT APPLICABLE ---
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 19

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	Tool calling: how models request tools	3	Talk	--- NOT APPLICABLE ---
20	Validated arguments and allow-lists	5	Talk	Quiz/Test Questions
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 20

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	Rubrics, regression tests, LLM-as-a-judge with human agreement	5	Talk	--- NOT APPLICABLE ---
20	Prompt injection, leakage, quality regression	6	PPT	Role playing
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 21

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	Fine-tuning vs prompting vs RAG	2	Talk	--- NOT APPLICABLE ---
20	Domain, style, narrow-task adaptation; build vs buy	2	PPT	--- NOT APPLICABLE ---
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 22

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	Full fine-tuning vs PEFT	2	Talk	--- NOT APPLICABLE ---
20	LoRA, QLoRA, adapters, quantisation	3	Talk	--- NOT APPLICABLE ---
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 23

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	Instruction dataset design and quality	2	Talk	--- NOT APPLICABLE ---
20	Dataset documentation; supervised fine-tuning	3	PPT	One-Minute Paper
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 24

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	RLHF and PPO	3	Talk	--- NOT APPLICABLE ---
20	DPO and RLAIF awareness	4	PPT	Quiz/Test Questions
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 25

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	Quality, effort, memory, latency, cost	4	Talk	--- NOT APPLICABLE ---
20	One test set for all options; learned router and decision record	5	PPT	Debate
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 26

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	Layers: UI, API, orchestration, retrieval, model, post-processing, safety	2	Talk	--- NOT APPLICABLE ---
20	Clean architecture, 12-factor config, model selection	2	PPT	--- NOT APPLICABLE ---
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 27

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	FastAPI microservices with Pydantic contracts	2	Talk	--- NOT APPLICABLE ---
20	React + Vite (TS/JS); Tailwind, MUI, Bootstrap	3	Talk	Video synthesis
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 28

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	One trace per request: Langfuse or LangSmith	4	Talk	--- NOT APPLICABLE ---
20	Cost control and graceful degradation	5	Talk	--- NOT APPLICABLE ---
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 29

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	Guardrails and human-in-the-loop	4	Talk	--- NOT APPLICABLE ---
20	Kill switch; ethics and personalisation governance	5	PPT	Quiz/Test Questions
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 30

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Talk	--- NOT APPLICABLE ---
20	Docker, docker-compose, hardened images; GitHub Actions with Trivy, tfsec/checkov	5	Talk	--- NOT APPLICABLE ---
20	Terraform on AWS (VPC, IAM, ECR, EC2); MCP as enrichment	6	Talk	--- NOT APPLICABLE ---
5	Summary and discussion	1	Talk	--- NOT APPLICABLE ---

Tutorial Course DELIVERY Plan:
NO Delivery Plan Exists

Tutorial Session wise Teaching – Learning Plan

No Session Plans Exists

Practical Course DELIVERY Plan:
NO Delivery Plan Exists

Practical Session wise Teaching – Learning Plan

No Session Plans Exists

Skilling Course DELIVERY Plan:

Skilling session no	Topics/Experiments	CO-Mapping
1	Set up the development environment: Python, virtual environment, Git and GitHub repository, VS Code, API keys in .env (12-factor); first LLM call	CO1
2	Build a provider-agnostic LLM client: one interface over a hosted API and a local Ollama model	CO1
3	Log tokens, latency and cost for every call to MongoDB; add timeouts and retries	CO1
4	Tokenizer lab: compare BPE, WordPiece and SentencePiece tokenizers on English, Telugu and Hindi text	CO1
5	Token economics: cost and rate-limit budget for a feature before building it	CO1
6	Write the golden question set with expected answers and sources; first baseline run — release v0.1	CO1

Skilling session no	Topics/Experiments	CO-Mapping
7	Next-token probabilities: inspect logits and top-k candidates from a small open model with Hugging Face Transformers	CO2
8	Implement scaled dot-product and multi-head attention from scratch in PyTorch	CO2
9	Visualise real attention weights in BERT and GPT-2	CO2
10	Compare encoder (BERT fill-mask), decoder (GPT-2 generation) and encoder–decoder (Flan-T5) models on the same tasks	CO2
11	Sampling sweep over temperature, top-k and top-p on the golden set; classify failures	CO2
12	Select generation defaults from measured evidence; write ADR-001 — release v0.2	CO2
13	Embeddings lab: sentence-transformers vs TF-IDF vs BM25 on course-material queries	CO3
14	Ingestion pipeline: load, clean, add metadata and chunk with overlap; store documents in MongoDB	CO3
15	Index embeddings in Chroma; measure HNSW recall against latency	CO3
16	Measure retrieval with Hit Rate@k and MRR; choose chunk size and k by measurement	CO3
17	Build the RAG endpoint: context construction, grounded generation and citations	CO3
18	Add the refusal path for unsupported questions; RAG evaluation report — release v0.3	CO3
19	Version prompts as files in a prompt registry; separate system and user prompts	CO4
20	Compare zero-shot, few-shot and role prompting with numbers on the golden set	CO4
21	Structured outputs with Pydantic: validation, retry and safe fallback	CO4
22	Controlled tool calling with validated arguments and an allow-list	CO4
23	Build an injection-and-leakage test suite (direct and indirect injection, system-prompt leakage)	CO4
24	Calibrate an LLM-as-a-judge against human labels; run regression tests in GitHub Actions — release v0.4	CO4
25	Build and document an instruction dataset (dataset card); fix one shared test set	CO5
26	Prompting and RAG baselines on the shared test set: accuracy, latency and cost	CO5

Skilling session no	Topics/Experiments	CO-Mapping
27	LoRA supervised fine-tuning of a small open model with Hugging Face PEFT on a cloud GPU notebook	CO5
28	QLoRA with 4-bit quantisation; compare memory, latency and quality	CO5
29	Preference tuning demonstration with DPO (TRL)	CO5
30	Compare fine-tuning, prompting and RAG; train a learned router; decision record — release v0.5	CO5
31	FastAPI microservices with Pydantic contracts: gateway, RAG service and model service	CO6
32	React + Vite (TypeScript) chat interface with streaming answers and citations; Tailwind with MUI or Bootstrap	CO6
33	Hyper-personalisation: user profiles in MongoDB, consent and privacy controls, ethics layer	CO6
34	Tracing with Langfuse or LangSmith: one trace per request, cost dashboard, graceful degradation	CO6
35	Guardrails, human-in-the-loop review queue and kill switch	CO6
36	Docker and docker-compose with hardened images; GitHub Actions with Trivy and tfsec/checkov	CO6
37	Terraform on AWS: VPC, IAM, ECR, EC2; deploy the stack	CO6
38	Expose course tools and resources through an MCP server as enrichment	CO6
39	Project Phase-I: integration, load and security testing; release v1.0	CO6
40	Project Phase-II: final demonstration and defence of the v0.1 → v1.0 record	CO6

Skilling Session wise Teaching – Learning Plan

SESSION NUMBER : 1

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Tooling set-up and repository layout	2	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
40	First LLM call from a script	2	LTC	--- NOT APPLICABLE ---
10	Documenting Results: README with set-up steps	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 2

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Client interface design	2	PPT	--- NOT APPLICABLE ---
40	Implement two providers behind one interface	3	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Client module + unit test	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 3

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Call logging schema in MongoDB	2	PPT	--- NOT APPLICABLE ---
40	Log tokens, latency, cost; add retries	3	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Sample call log	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 4

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Tokenizer libraries walkthrough	2	PPT	--- NOT APPLICABLE ---
40	Token counts for English, Telugu, Hindi	3	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Token-ratio table	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 5

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Cost model for a feature	2	PPT	--- NOT APPLICABLE ---
40	Budget sheet for 1,000 daily users	3	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Cost budget note	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 6

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Golden-set format and scoring	2	PPT	--- NOT APPLICABLE ---
40	Write 30+ questions; baseline run	3	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Release v0.1 + baseline scores	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 7

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Reading logits and probabilities	2	PPT	--- NOT APPLICABLE ---
40	Top-k candidates for prompts	2	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Probability table	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 8

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Attention as matrix operations	2	PPT	--- NOT APPLICABLE ---
40	Code single- and multi-head attention	3	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Attention module + test	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 9

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Attention visualisation tools	3	PPT	--- NOT APPLICABLE ---
40	Inspect heads in BERT and GPT-2	4	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Annotated attention plots	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 10

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Three families on one task set	3	PPT	--- NOT APPLICABLE ---
40	Run BERT, GPT-2, Flan-T5	4	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Comparison table	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 11

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Sweep design and failure taxonomy	3	PPT	--- NOT APPLICABLE ---
40	Sweep temperature, top-k, top-p	4	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Failure classification sheet	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 12

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	ADR template	3	PPT	--- NOT APPLICABLE ---
40	Choose defaults from evidence	4	LTC	--- NOT APPLICABLE ---
10	Documenting Results: ADR-001 + release v0.2	1	Talk	--- NOT APPLICABLE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
				---

SESSION NUMBER : 13

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Dense vs sparse retrieval	2	PPT	--- NOT APPLICABLE ---
40	Compare three retrievers	2	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Retriever comparison	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 14

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Ingestion pipeline design	2	PPT	--- NOT APPLICABLE ---
40	Clean, add metadata, chunk, store	3	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Ingestion script + MongoDB collection	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 15

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Chroma collections and HNSW settings	3	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
40	Index and time queries	3	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Recall–latency table	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 16

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Hit Rate@k and MRR harness	4	PPT	--- NOT APPLICABLE ---
40	Tune chunk size and k	5	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Retrieval metrics report	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 17

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	RAG endpoint with citations	5	PPT	--- NOT APPLICABLE ---
40	Grounded answers on the golden set	6	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Cited answers log	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 18

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Refusal logic for unsupported questions	5	PPT	--- NOT APPLICABLE ---
40	Test out-of-scope questions	6	LTC	--- NOT APPLICABLE ---
10	Documenting Results: RAG report + release v0.3	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 19

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Prompt registry and versioning	2	PPT	--- NOT APPLICABLE ---
40	Move prompts to versioned files	3	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Prompt registry	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 20

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Prompt experiment harness	4	PPT	--- NOT APPLICABLE ---
40	Zero- vs few-shot vs role	5	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Scored comparison	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 21

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Pydantic output schemas	2	PPT	--- NOT APPLICABLE ---
40	Validation, retry, fallback	3	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Schema module + tests	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 22

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Tool-calling loop	5	PPT	--- NOT APPLICABLE ---
40	Validated tools with allow-list	6	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Tool-call log	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 23

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Attack catalogue	4	PPT	--- NOT APPLICABLE ---
40	Write injection and leakage tests	5	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Security test suite	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 24

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Judge calibration and CI	5	PPT	--- NOT APPLICABLE ---
40	Human labels vs judge; CI workflow	6	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Agreement score + release v0.4	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 25

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Instruction data and dataset cards	2	PPT	--- NOT APPLICABLE ---
40	Build and split dataset	3	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Dataset + dataset card	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 26

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Baseline protocol	3	PPT	--- NOT APPLICABLE ---
40	Run prompting and RAG baselines	4	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Baseline table	1	Talk	--- NOT APPLICABLE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
				---

SESSION NUMBER : 27

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	PEFT and LoRA configuration	2	PPT	--- NOT APPLICABLE ---
40	Fine-tune on a GPU notebook	3	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Adapter + training log	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 28

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	4-bit quantisation with QLoRA	3	PPT	--- NOT APPLICABLE ---
40	Measure memory and latency	4	LTC	--- NOT APPLICABLE ---
10	Documenting Results: QLoRA comparison	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 29

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	DPO with TRL	3	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
40	Run a small preference-tuning job	4	LTC	--- NOT APPLICABLE ---
10	Documenting Results: DPO observations	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 30

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Router and decision record	4	PPT	--- NOT APPLICABLE ---
40	Train router; compare all options	5	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Decision record + release v0.5	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 31

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Service boundaries and contracts	2	PPT	--- NOT APPLICABLE ---
40	Build gateway, RAG and model services	3	LTC	--- NOT APPLICABLE ---
10	Documenting Results: OpenAPI docs	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 32

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	React + Vite project structure	2	PPT	--- NOT APPLICABLE ---
40	Chat UI with streaming and citations	3	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Working front end	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 33

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Profiles, consent and privacy	5	PPT	--- NOT APPLICABLE ---
40	Personalised answers with opt-out	6	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Ethics-layer note	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 34

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Tracing set-up	4	PPT	--- NOT APPLICABLE ---
40	One trace per request; fallbacks	5	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Trace screenshots + cost view	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 35

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Guardrails and review queue	4	PPT	--- NOT APPLICABLE ---
40	Kill switch drill	5	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Guardrail test results	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 36

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Dockerfiles and CI scans	5	PPT	--- NOT APPLICABLE ---
40	Hardened images; Trivy, tfsec/checkov in CI	6	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Passing pipeline	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 37

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Terraform modules for AWS	5	PPT	--- NOT APPLICABLE ---
40	Plan and apply VPC, IAM, ECR, EC2	6	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Deployed URL	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 38

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	MCP server basics	5	Talk	--- NOT APPLICABLE ---
40	Expose course tools and resources	6	Talk	--- NOT APPLICABLE ---
10	Documenting Results: MCP demo	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 39

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Test plan for v1.0	5	PPT	--- NOT APPLICABLE ---
40	Integration, load and security tests	6	LTC	--- NOT APPLICABLE ---
10	Documenting Results: Release v1.0 notes	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 40

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Introduction about the Experiment	1	Talk	--- NOT APPLICABLE ---
40	Demonstration format	5	PPT	--- NOT APPLICABLE ---
40	Final demo and viva	6	LTC	--- NOT APPLICABLE ---
10	Documenting Results: v0.1 → v1.0 record	1	Talk	--- NOT APPLICABLE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
				---

WEEKLY HOMEWORK ASSIGNMENTS/ PROBLEM SETS/OPEN ENDEDED PROBLEM-SOLVING EXERCISES etc:

Week	Assignment Type	Assignment No	Topic	Details	co

COURSE TIME TABLE:

	Hour	1	2	3	4	5	6	7	8	9
Day	Component									
Mon	Theory	--	--	--	--	--	---	--	---	---
	Tutorial	--	--	--	--	--	---	--	---	---
	Lab	--	--	--	--	--	---	--	---	---
	Skilling	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6,H-S10,H-S10,H-S11,H-S11,H-S14,H-S14	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6,H-S10,H-S10,H-S11,H-S11,H-S14,H-S14	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9,H-S15,H-S15	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9,H-S15,H-S15	H-S12,H-S12,H-S13,H-S13	---	H-S12,H-S12,H-S13,H-S13	---	---
Tue	Theory	H-S7,H-S8,H-S9,H-S12,H-S13	H-S7,H-S8,H-S9,H-S12,H-S13	--	--	--	---	--	---	---
	Tutorial	--	--	--	--	--	---	--	---	---
	Lab	--	--	--	--	--	---	--	---	---
	Skilling	H-S15,H-S15	H-S15,H-S15	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6,H-S10,H-S10,H-S11,H-S11	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6,H-S10,H-S10,H-S11,H-S11	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9,H-S12,H-S12,H-S13,H-S13,H-S14,H-S14	---	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9,H-S12,H-S12,H-S13,H-S13,H-S14,H-S14	---	---
Wed	Theory	H-S1,H-S2,H-S3	H-S1,H-S2,H-S3	H-S15	H-S15	--	---	--	---	---
	Tutorial	--	--	--	--	--	---	--	---	---
	Lab	--	--	--	--	--	---	--	---	---
	Skilling	--	--	--	--	H-S1,H-S1,H-S2,H-	---	H-S1,H-S1,H-S2,H-	---	---

	Hour	1	2	3	4	5	6	7	8	9
						S2,H-S3,H-S3		S2,H-S3,H-S3		
Thu	Theory	--	--	H-S4,H-S5,H-S6,H-S14	H-S4,H-S5,H-S6,H-S14	--	---	--	---	---
	Tutorial	--	--	--	--	--	---	--	---	---
	Lab	--	--	--	--	--	---	--	---	---
	Skilling	H-S1,H-S1,H-S2,H-S2,H-S3,H-S3	H-S1,H-S1,H-S2,H-S2,H-S3,H-S3	H-S10,H-S10,H-S11,H-S11	H-S10,H-S10,H-S11,H-S11	H-S1,H-S1,H-S2,H-S2,H-S3,H-S3	---	H-S1,H-S1,H-S2,H-S2,H-S3,H-S3	---	---
Fri	Theory	--	--	H-S10,H-S11	H-S10,H-S11	--	---	--	---	---
	Tutorial	--	--	--	--	--	---	--	---	---
	Lab	--	--	--	--	--	---	--	---	---
	Skilling	H-S14,H-S14	H-S14,H-S14	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9,H-S12,H-S12,H-S13,H-S13,H-S15,H-S15	---	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9,H-S12,H-S12,H-S13,H-S13,H-S15,H-S15	---	---
Sat	Theory	--	--	--	--	--	---	--	---	---
	Tutorial	--	--	--	--	--	---	--	---	---
	Lab	--	--	--	--	--	---	--	---	---
	Skilling	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9,H-S12,H-S12,H-S13,H-S13,H-S15,H-S15	H-S7,H-S7,H-S8,H-S8,H-S9,H-S9,H-S12,H-S12,H-S13,H-S13,H-S15,H-S15	H-S1,H-S1,H-S2,H-S2,H-S3,H-S3	H-S1,H-S1,H-S2,H-S2,H-S3,H-S3	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6,H-S10,H-S10,H-S11,H-S11,H-S14,H-S14	---	H-S4,H-S4,H-S5,H-S5,H-S6,H-S6,H-S10,H-S10,H-S11,H-S11,H-S14,H-S14	---	---
Sun	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	--	--	--	--	--	--	--	--	--

REMEDIAL CLASSES:

Supplementary course handouts, including special lectures and discussions, will be planned and the schedule will be notified accordingly.

SELF-LEARNING:

Assignments to promote self-learning, survey of contents from multiple sources.

S.no	Topics	CO	ALM	References/MOOCS

DELIVERY DETAILS OF CONTENT BEYOND SYLLABUS:

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.no	Advanced Topics, Additional Reading, Research papers and any	CO	ALM	References/MOOCS

EVALUATION PLAN:

Evaluation Type	Evaluation Component	Weightage/Marks		Assessment Dates	Duration (Minutes)	CO1	CO2	CO3	CO4	CO5	CO6
End Semester Summative Evaluation Total= 40 %	Term End Hackathon	Weightage	16		180	2	2	2	3	3	4
		Max Marks	90			15	15	15	15	15	15
	End Semester Exam	Weightage	24	1-2-2027,1-2-2027,1-2-2027,1-2-2027,1-2-2027	180	2	2	2	6	6	6
		Max Marks	90			5	5	5	25	25	25
In Semester Formative Evaluation Total= 36 %	AI-Enabled Adaptive Learning Engagement	Weightage	20		100	3.3	3.3	3.3	3.3	3.3	3.5
		Max Marks	60			10	10	10	10	10	10
	Project Continuous Evaluation	Weightage	8		100	1.3	1.3	1.3	1.3	1.3	1.5
		Max Marks	60			10	10	10	10	10	10
	Skilling Continuous Evaluation	Weightage	8		100	1.3	1.3	1.3	1.3	1.3	1.5
		Max Marks	60			10	10	10	10	10	10
In Semester Summative Evaluation Total= 24 %	MOOCs Completion & Certification	Weightage	6		100	1	1	1	1	1	1
		Max Marks	60			10	10	10	10	10	10
	In-Term Practical Examination	Weightage	6		100	1	1	1	1	1	1
		Max Marks	60			10	10	10	10	10	10
	Mid-Term Examination (Descriptive)	Weightage	12	14-12-2026,14-12-2026,14-12-2026	90	4	4	4			
		Max Marks	45			15	15	15			

ATTENDANCE POLICY:

Every student is expected to be responsible for regularity of his/her attendance in class rooms and laboratories, to appear in scheduled tests and examinations and fulfill all other tasks assigned to him/her in every course
In every course, student has to maintain a minimum of 85% attendance to be eligible for appearing in Semester end examination of the course, for cases of medical issues and other unavoidable circumstances the students will be condoned if their attendance is between 75% to 85% in every course, subjected to submission of medical certificates, medical case file and other needful documental proof to the concerned departments

DETENTION POLICY :

In any course, a student has to maintain a minimum of 85% attendance and In-Semester Examinations to be eligible for appearing to the Semester End Examination, failing to fulfill these conditions will deem such student to have been detained in that course.

PLAGIARISM POLICY :

Plagiarism is treated as a serious academic offense, and the university enforces a strict zero-tolerance policy to uphold academic integrity. For a first offense, minor cases may result in a warning and reduced marks, while major cases may lead to a zero grade and mandatory academic integrity training. A second offense will result in a zero grade, formal reprimand, and academic probation, whereas a third offense may lead to suspension and a permanent record of misconduct. Repeated violations can result in expulsion, and in group work, all members are held accountable unless responsibility is clearly established.

COURSE TEAM MEMBERS, CHAMBER CONSULTATION HOURS AND CHAMBER VENUE DETAILS:

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Dyuti Banerjee	L	13-MA	-	-	-	-
Dyuti Banerjee	S	13-A	-	-	-	-
Gunturi Raghavendra	L	1-MA	-	-	-	-
Gunturi Raghavendra	S	5-B,1-A	-	-	-	-
ASIF SHAIK	L	8-MA	-	-	-	-
ASIF SHAIK	S	5-B,8-A	-	-	-	-
khaja sk	L	11-MA	-	-	-	-
khaja sk	S	11-A	-	-	-	-
Anitha Chekuri	S	12-B	-	-	-	-
Nirmalajyothi Narisetty	L	2-MA	-	-	-	-
Nirmalajyothi Narisetty	S	9-B,2-A	-	-	-	-
LakshmiPrasanna Yeluri	L	10-MA	-	-	-	-
LakshmiPrasanna Yeluri	S	10-A,12-B	-	-	-	-
Swathi D	S	13-B	-	-	-	-
KIRAN KUMAR ANUMANDLA	L	14-MA	-	-	-	-
KIRAN KUMAR ANUMANDLA	S	15-B,14-A	-	-	-	-
Chandusha kanda	S	13-B,10-B,12-B	-	-	-	-
Purushotham Muniganti	L	5-MA	-	-	-	-
Purushotham Muniganti	S	5-A,8-B	-	-	-	-
Jyothsna Datti	L	4-MA	-	-	-	-

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Jyothsna Datti	S	3-B,4-A,7-B	-	-	-	-
Rahul LR	L	15-MA	-	-	-	-
Rahul LR	S	14-B,15-A	-	-	-	-
I V SAI HARITHA	L	12-MA	-	-	-	-
I V SAI HARITHA	S	12-A,10-B	-	-	-	-
Kavitha Matta	L	9-MA	-	-	-	-
Kavitha Matta	S	9-A,6-B,2-B	-	-	-	-
Gayatri Mantri	S	11-B	-	-	-	-
Kotte Anusha	L	7-MA	-	-	-	-
Kotte Anusha	S	7-A,6-B,2-B	-	-	-	-
DHARAMSOTU BHEEKYA	L	6-MA	-	-	-	-
DHARAMSOTU BHEEKYA	S	1-B,7-B,6-A	-	-	-	-
pavani Chilumula	S	4-B	-	-	-	-
Pratima Depa	S	1-B	-	-	-	-
Divya Priya Degala	S	11-B	-	-	-	-
MEKALA SANJEEV KUMAR	S	11-B	-	-	-	-
Anu Ragavi V	L	3-MA	-	-	-	-
Anu Ragavi V	S	3-A,7-B,4-B	-	-	-	-

GENERAL INSTRUCTIONS

Students should come prepared for classes by reviewing the prescribed video lectures in advance and carrying the required textbook(s) or study material(s) as specified by the Course Faculty.

NOTICES

All course-related notices and communications will be shared through the official institutional email.

Signature of COURSE COORDINATOR

(DHARAMSOTU BHEEKYA)(10006)(CSE-1)

Signature of Department Prof. Incharge Academics & Vetting Team Member

Department Of CSE-1

HEAD OF DEPARTMENT:

Approval from: DEAN-ACADEMICS
(Sign with Office Seal)
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